Bureau
International des
Poids et
Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Iodine ()

Iodine ()

CCL-CCTF Frequency Standards Working Group



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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Iodine ()

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Appendix 2

1. CIPM recommended values

The values $f = 489\ 880\ 354.9\ \text{MHz}$
$\lambda = 611\ 970\ 770.0\ \text{fm}$

with a relative standard uncertainty of 3×10^{-10} apply to the radiation of a He-Ne laser stabilized with an iodine cell, within or external to the laser, having a cold-finger temperature of $(-5 \pm 2)\ °C$[(1)].

[(1)] For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols ± refer to a tolerance, not an uncertainty.

2. Source data

Adopted value

$f = 489\ 880\ 354.93\ (15)$ MHz

$u_c/y = 3.0 \times 10^{-10}$

for which:

$\lambda = 611\ 970\ 769.97\ (18)$ fm

$u_c/y = 3.0 \times 10^{-10}$

calculated from

f/ kHz	u_c/y	source data
489 880 354 979	1×10^{-10}	Section 2.1
489 880 354 728	2.1×10^{-10}	Section 2.2
489 880 355 026	8.3×10^{-11}	Section 2.3
489 880 355 062	3.0×10^{-10}	Section 2.4
489 880 358 850	8.5×10^{-11}	Section 2.5
Unweighted mean:	$f = 489\ 880\ 354\ 929$ kHz	

Other available values having relative uncertainties higher than 3.0×10^{-10} have not been used. The relative standard uncertainty calculated from the dispersion of the six values is 2.8×10^{-10}, which the CCL preferred to round up to 3.0×10^{-10}.

Source data

2.1.

Reference [1] gives

$f_{a7}/f_i = 1.034\ 349\ 072\ 43$

$u_c/y = 1 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, a_{16} or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_i = 473\ 612\ 214\ 712$ kHz

$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a7} = 489\ 880\ 354\ 979$ kHz

$u_c/y = 1 \times 10^{-10}$.

2.2.

Reference [2] gives

$f_{a7}/f_i = 1.034\ 349\ 071\ 90$

$u_c/y = 2.1 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, a_{16} or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_i = 473\ 612\ 214\ 712$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a7} = 489\ 880\ 354\ 728$ kHz
$u_c/y = 2.1 \times 10^{-10}$.

2.3.

Bönsch et al. [3] give $\lambda_{b15}/\lambda_i = 0.966\ 791\ 921\ 43$
$u_c/y = 8 \times 10^{-11}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, a_{16} or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_i = 473\ 612\ 214\ 712$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{b15} = 489\ 880\ 194\ 708$ kHz
$u_c/y = 8.3 \times 10^{-11}$.

From the measured value (see Table 4 below)
$f_{b15} - f_{a7} = -160\ 318$ kHz
$u_c = 3$ kHz

one calculates

$f_{a7} = 489\ 880\ 355\ 026$ kHz
$u_c/y = 8.3 \times 10^{-11}$.

2.4.

Vitushkin et al. [4] give $\lambda_d/\lambda_{a7} = 1.034\ 348\ 712$
$u_c/y = 3 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, a_{16} or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_d = 473\ 612\ 379\ 828$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a7} = 489\ 880\ 355\ 062$ kHz

$u_c/y = 3.0 \times 10^{-10}$.

2.5.

Himbert et al. [5] give $f_{a13} = 489\ 880\ 604\ 541$ kHz
$u_c = 88$ kHz.

This value is a result of the frequency ratio f_{a13}/f_e, to which the recommended value adopted by the CIPM in 1983 [6], [7] was applied, i.e. $f_i = 473\ 612\ 214.8$ MHz. (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8)

$f_e - f_i = 152\ 255$ kHz
$u_c = 5$ kHz,

one obtains

$f_e = 473\ 612\ 367\ 055$ kHz,

and hence

$f_{a13}/f_e = 1.034\ 349\ 267$
$u_c/y = 8 \times 10^{-11}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, a_{16} or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_e = 473\ 612\ 366\ 967$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a13} = 489\ 880\ 604\ 450$
$u_c/y = 8.3 \times 10^{-11}$.

Knowing the frequency difference (see Table 1, see p. 8)
$f_{a7} - f_{a13} = -249\ 600$ kHz
$u_c = 10$ kHz,

one obtains

$f_{a7} = 489\ 880\ 354\ 850$
$u_c/y = 8.5 \times 10^{-11}$.

3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 184-187 and *Metrologia*, 2003, **40**, 127-128.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, u_c represents the estimated combined standard uncertainty (1σ).

All transitions in molecular iodine refer to the B-X system.

Table 1

$\lambda \approx 612$ nm $^{127}I_2$ R(47) 9-2

a_n	x	$[f(a_n) - f(a_7)]$/ MHz	u_c/ MHz	a_n	x	$[f(a_n) - f(a_7)]$/ MHz	u_c/ MHz
a_1	u	-357.16	0.02	a_{12}	j	219.602	0.006
a_2	t	-333.97	0.01	a_{13}	i	249.60	0.01
a_3	s	-312.46	0.02	a_{14}	h	284.30	0.01
a_4	r	-86.168	0.007	a_{15}	g	358.37	0.03
a_5	q	-47.274	0.004	a_{16}	f	384.66	0.01
a_6	p	-36.773	0.003	a_{17}	e	403.76	0.02
a_7	o	0	—	a_{18}	d	429.99	0.02
a_8	n	81.452	0.003	a_{19}	c	527.16	0.02
a_9	m	99.103	0.003	a_{20}	b	539.22	0.02
a_{10}	l	107.463	0.005	a_{21}	a	555.09	0.02
a_{11}	k	119.045	0.006				

Frequency referenced to: a_7: R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [9], [10], [11], [12], [13], [14]

Table 2

$\lambda \approx 612$ nm $^{127}I_2$ P(48) 11-3

b_n	$[f(b_n)-f(a_7)]$/MHz	u_c/ MHz	b_n	$[f(b_n)-f(a_7)]$/MHz	u_c/ MHz
b_1	-1 034.75	0.07	b_9	-579.91	0.01
b_2	-755.86	0.05	b_{10}	-452.163	0.005
b_3	-748.28	0.03	b_{11}	-316.6	0.4
b_4	-738.35	0.04	b_{12}	-315.8	0.4
b_5	-731.396	0.006	b_{13}	-297.42	0.03
b_6	-616.01	0.03	b_{14}	-294.72	0.03
b_7	-602.42	0.03	b_{15}	-160.318	0.003
b_8	-593.98	0.01			

Frequency referenced to a_7, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [9], [10], [12], [13], [14], [15]

Table 3

$\lambda \approx 612$ nm $^{127}I_2$ R(48) 15-5

c_n	$[f(c_n)-f(a_7)]$/MHz	u_c/ MHz	c_n	$[f(c_n)-f(a_7)]$/MHz	u_c/ MHz
c_1	-513.83	0.03	c_5	-209.96	0.03
c_2	-237.40	0.03	c_6	-97.74	0.03
c_3	-228.08	0.03	c_8	-73.92	0.03
c_4	-218.78	0.03	c_9	-59.30	0.03

Frequency referenced to a_7, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [10]

Table 4

$\lambda \approx 612$ nm $^{129}I_2$ P(110) 10-2

a_n	x	$\left[f(a_n)-f\left(a_7\{^{127}I_2\}\right)\right]$ / MHz	u_c/ MHz	a_n	x	$\left[f(a_n)-f\left(a_7\{^{127}I_2\}\right)\right]$ / MHz	u_c/ MHz
a_1	b’	-376.29	0.05	a_{15}	n	1.61	0.20
a_2	a’	-244.76	0.10	a_{16}	m	10.63	0.15
a_3	z	-230.79	0.20	a_{17}	l	15.82	0.20
a_4	y	-229.40	0.20	a_{18}	k	25.32	0.10
a_5	x	-216.10	0.05	a_{19}	j	49.44	0.15
a_6	w	-149.37	0.10	a_{20}	i	54.66	0.20
a_7	v	-134.68	0.10	a_{21}	h	69.02	0.10
a_8	u	-130.98	0.10	a_{22}	g	74.47	0.15
a_9	t	-116.67	0.05	a_{23}	f	110.60	0.10
a_{10}	s	-96.26	0.20	a_{24}	e	153.09	0.20

a_{11}	r	-90.70	0.20	a_{25}	d	154.70	0.20
a_{12}	q	-84.12	0.20	a_{26}	c	163.98	0.20
a_{13}	p	-77.79	0.20	a_{27}	b	166.22	0.20
a_{14}	o	-72.70	0.20	a_{28}	a	208.29	0.10

Frequency referenced to R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [16], [17], [18]

Table 5

$\lambda \approx 612$ nm $^{129}I_2$ R(113) 14-4

b_n	x	$[f(b_n) - f(a_7\{^{127}I_2\})]$ / MHz	u_c / MHz	b_n	x	$[f(b_n) - f(a_7\{^{127}I_2\})]$ / MHz	u_c / MHz
b_{19}	r	-410.4	0.3	b_{28}	i	-289.4	0.5
b_{20}	q	-390.0	0.3	b_{29}	h	-273.1	0.3
b_{21}	p	-383.9	0.5	b_{30}	g	-255.7	0.5
b_{22}	o	-362.8	0.3	b_{31}	f	-247	5
b_{23}	n	-352.9	0.3	b_{32}	e	-237	5
b_{24}	m	-346.4	0.3	b_{33}	d	-223	5
b_{25}	l	-330.0	0.3	b_{34}	c	-198.6	0.3
b_{26}	k	-324.9	0.3	b_{35}	b	-193.1	0.3
b_{27}	j	-304.7	0.3	b_{36}	a	-187.0	0.3

Frequency referenced to R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [17], [18]

Absorbing molecule $^{127}I_2$, a_7 component, R(47) 9-2 transition[1]

[1] All transitions in I_2 refer to the $B^3\Pi\, 0_u^+ - X^1\, \Sigma_g^+$ system.

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